AERM Exam Outputs and Results

2026-03-23

Part 1

Use Julia code

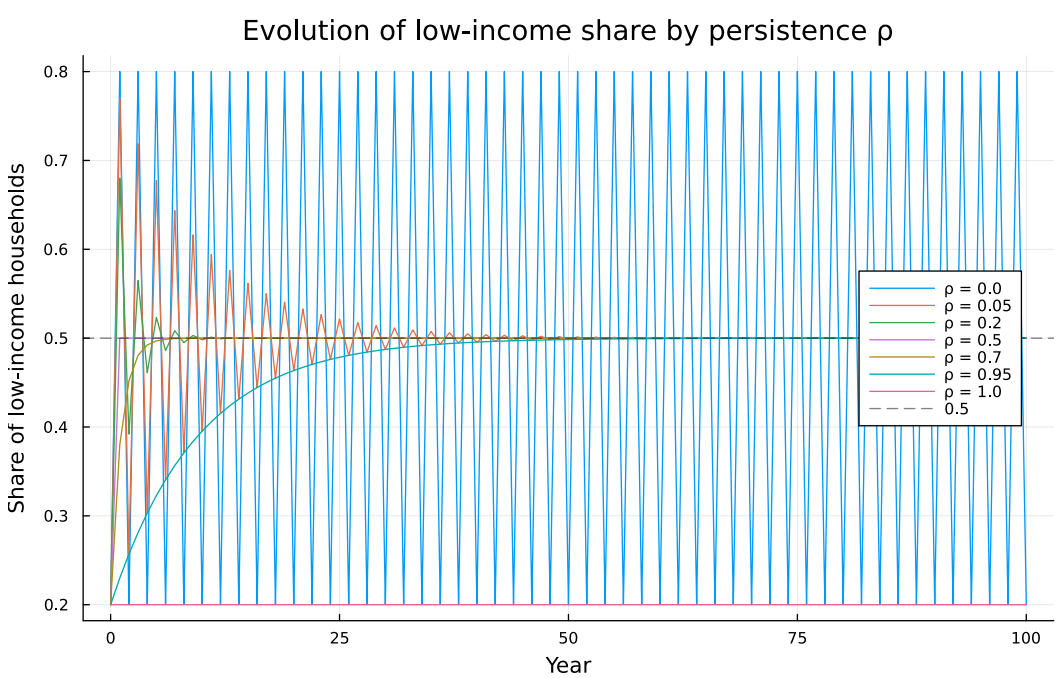
Exercise 1: Solve Standard Consumption-Saving Problem for Different Values of ρ

Question 1

Income follows a two-state Markov chain with symmetric persistence $\rho_L = \rho_H = \rho$ and transition matrix:

$$P = \begin{pmatrix} \rho & 1-\rho \\ 1-\rho & \rho \end{pmatrix}$$

Starting from $P(y = y_L) = 0.2$, the figure below shows the share of low-income households over a 100-year horizon for each value of ρ.



Explanation:

For all $\rho \in (0, 1)$, the share of low-income households converges to 0.5, which is the unique stationary distribution of this symmetric Markov chain, but higher persistence slows the rate of convergence.

The two edge cases behave differently: $\rho = 0$ causes full state-switching each period (so the share oscillates before settling), while $\rho = 1$ makes both states absorbing, so the initial distribution $P(y_L) = 0.2$ is permanent.

Question 2

ρ	$\mathrm{Var}(\Delta \log y)$
0.00	0.48045
0.05	0.45643
0.20	0.38436
0.50	0.24023
0.70	0.14414
0.95	0.02402
1.00	0.00000

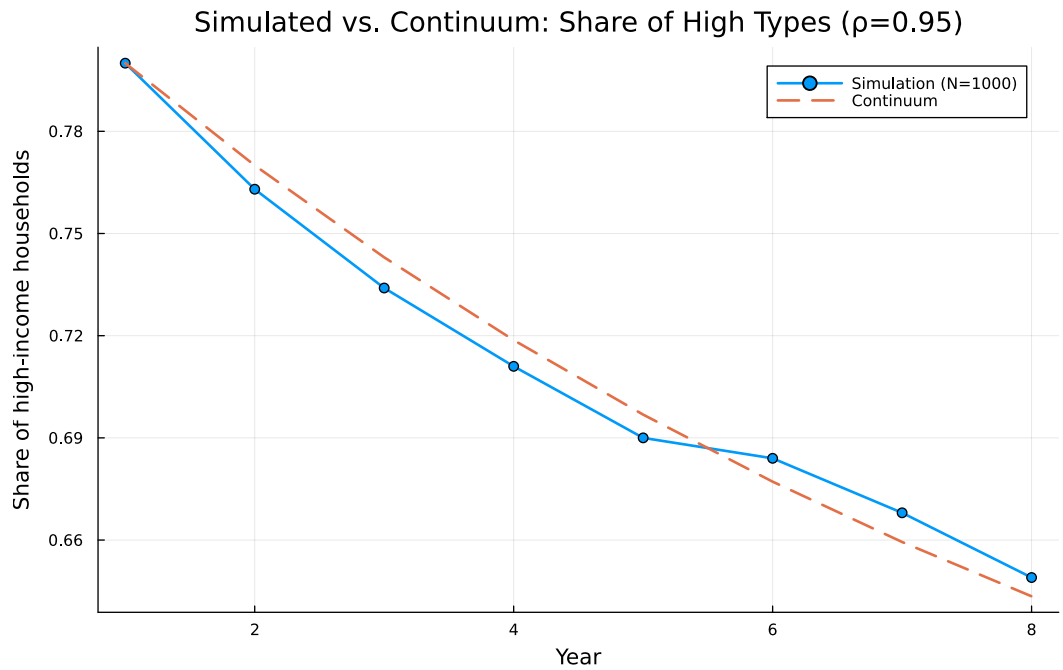
Labour income volatility, measured as $\mathrm{Var}(\Delta \log y)$, is maximised at $\rho = 0$ since households switch between income states every period, making changes in income most frequent. However, $\rho = 0$ does not generate the most risk: when income alternates deterministically, households can perfectly anticipate future income and smooth consumption via bond savings, whereas for intermediate or high values of ρ, income shocks are harder to insure against because the current state provides less — or too much persistent — information about future income.

Explanation:

In the stationary distribution, the variance of $\Delta \log y$ equals $(1 - \rho) \cdot (\log 2)^2$, which is maximized at $\rho = 0$ where every household switches state each period.

However, $\rho = 0$ does not generate the most risk for households: with i.i.d. income, bad shocks are transitory and easily self-insured through savings, whereas high persistence (e.g. $\rho = 0.95$) means adverse shocks last many periods, making them far harder to smooth.

Question 3



The simulated share of high-income types closely tracks the continuum prediction but does NOT match it exactly, because a finite sample of 1000 households is subject to sampling variation whereas the continuum result holds by the law of large numbers over a unit mass of agents.

Question 4

Using the simulated panel data, I regress current income on lagged income via OLS and fixed effects (implemented by demeaning), using only linear algebra as instructed.

Model	$\hat{\beta}$
OLS	0.8983
FE	0.5690
True ρ	0.95

OLS is closer to the true $\rho = 0.95$ because the data-generating process contains no individual fixed effects — all households face the same Markov chain — so OLS is consistent. The FE estimator suffers from Nickell bias: with only $T = 8$ periods, demeaning mechanically creates a negative correlation between the demeaned lagged variable and the demeaned error, biasing $\hat{\beta}_{FE}$ substantially downward. This bias is systematic and would shrink only as $T \to \infty$.